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## 1.0 EXECUTIVE SUMMARY

The proposed Manganese mining area of 126.84 Hectare (ha) in Gumgaon, Khodegaon and Tegai Village, Saoner Tehsil, Nagpur District, Maharashtra has been granted lease to M/s. MOIL Limited approved by Regional Controller, Nagpur Region, IBM vide letter no. NGP/MN/MPLN-1174/NGP-2016 on dated 03.10.2016.

The proposed manganese ore production is 1,02,000 Tonnes (TPA) ROM. The mining is underground mining. The region has good deposits of Manganese and has major demand in Steel Industry. The location advantage of the mine makes it possible to dispatch the Manganese in all the directions giving easy accessibility to the market.

**Table: Salient Features of the Project**

|                         |          |                                                                                                                                                                                                                     |
|-------------------------|----------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Name of the project     |          | Gumgaon, Khodegaon and Tegai Manganese Mine                                                                                                                                                                         |
| Promoter                |          | MOIL Limited                                                                                                                                                                                                        |
| Location                | Village  | Gumgaon, Khodegaon and Tegai                                                                                                                                                                                        |
|                         | Tehsil   | Saoner                                                                                                                                                                                                              |
|                         | District | Nagpur                                                                                                                                                                                                              |
|                         | State    | Maharashtra                                                                                                                                                                                                         |
| Khasra details          |          | Khasra Detail is enclosed as Annexure No. III                                                                                                                                                                       |
| Total Area              |          | 126.84 Ha.                                                                                                                                                                                                          |
| Water Requirement       |          | Ground water required through underground mining seepage (510 KLD) and bore well (210 KLD), total water requirement for various purpose will be 710 KLD.                                                            |
| Power Requirement       |          | M.S.E.B. 11 KV Line is provided up to village Khapa and near manganese deposit of Gumgaon area.                                                                                                                     |
| Manpower Requirement    |          | 456 workers                                                                                                                                                                                                         |
| Estimated Project cost  |          | Rs. 100 Crores                                                                                                                                                                                                      |
| Nearest Railway Station |          | Savner Railway Station (4.0 kms SW)                                                                                                                                                                                 |
| Nearest Highway         |          | <ul style="list-style-type: none"> <li>NH-69 - 5.20 km SW direction</li> <li>SH-249 – 0.70 km W direction</li> </ul>                                                                                                |
| Nearest Airport         |          | Nagpur 46.4 km S direction                                                                                                                                                                                          |
| Schooling facilities    |          | Schooling facilities are available at khapa Where Primary/High schools are available. Distance – 1.35 Km N direction                                                                                                |
| Hospital                |          | Company runs a Small Hospital at Gumgaon Mine which is 1 km from the lease area having two bed for male and two bed for female patients. Government hospital is available at Khapa (1.35 Km.) and Savner (4.12 Km). |

## 2.0 INTRODUCTION OF THE PROJECT/BACKGROUND INFORMATION

### (i) Identification of the Project

Govt. of India vide Gazette notification no 571, dated October 6, 2009 has reserved over an area of 814.71 Ha. in Bhandra and Nagpur district in favour of MOIL limited for exploitation of Manganese ore. Accordingly MOIL has applied PL within the reserved area. Govt. of Maharashtra vide letter no. PLS-0611/C.R. 19 /IND- 9 dated 1.7.2013 has granted prospecting license for Manganese ore over an area of 130.09 ha. MOIL has carried out geophysical prospecting followed by core drilling by 29 bore holes with 9370 mtrs of core drilling in the applied area during the years 2014-15 & 15-16 and applied ML application within 126.84 Ha. Details are enclosed in feasibility report. Two (2) blocks has been identified by exploration and it is proved that manganese ore is lying at 600'L up to 1300'L below the surface at Block I and 500'L up to 800'L in Block-II. Exploration has been conducted 3.25 ha revenue forest land. Since the forest land is devoid of any mineralization, it is excluded in ML area. This ML has been issued by Govt. of Maharashtra through PL route.

The Mines and Mineral (Development) Amendment Act 2015 came into force from 12.01.2015. The applied area is saved under 10A (2) of Mines and Mineral (Development) Amendment, Act 2015 and as the applied area is reserved under 17 (A) of MMDR act 1957 LOI of Central Govt. is not required. Copies of letter of Central Govt. addressed to Govt. of Maharashtra regarding LOI of MOIL's two other areas are attached as annexure III a & III B The LOI for ML area of 126.84 ha. has been issued on 6th April, 2016 through PL route.

The LOI for Mining Lease has been issued to MOIL over an area of 126.84 ha. in village Gumgaon, Kodegaon and Tegai, Tah.: Saoner of Dist: Nagpur of Maharashtra State, by Government of Maharashtra vide letter number MMN-0216/L. No. 20/Industry-9, Mumbai dated 06.04.2016. Copy of the LOI and Khasra Schedule is enclosed as **Annexure I & III**. Approval of Mining plan in respect of Manganese ore deposit of M/s Moil Ltd. is vide letter no. NGP/MN/MPLN-1174/NGP-2016 on dated 03.10.2016, over an area of 126.84 ha. situated in village- Gumgaon, Kodegaon & Tegai, Tehsil Saoner, District Nagpur of Maharashtra State is enclosed as **Annexure II**

### (ii) Project Proponent

A Miniratna PSU was originally set up in the year 1896 as Central Province Prospecting Syndicate which was later renamed as Central Provinces Manganese Ore Company Limited (CPMO), a British Company incorporated in the UK. In 1962, as a result of an agreement between the Government of India and CPMO, the assets of the latter were taken over by the Government and MOIL was formed with 51% capital held between the Govt. of India and the State Governments of Maharashtra and Madhya Pradesh and the balance 49% by CPMO. It was in 1977, the balance 49% shareholding was acquired from

CPMO and MOIL became a 100% Government Company under the administrative control of the Ministry of Steel.

At present, MOIL operates 10 mines, six located in the Nagpur and Bhandara districts of Maharashtra and four in the Balaghat district of Madhya Pradesh. All these mines are about a century old. Except 3, rest of the mines are worked through underground method. The Balaghat Mine is the largest mine of the Company. The mine has now reached a mining depth of 309 meters from the surface. Dongri Buzurg Mine located in the Bhandara district of Maharashtra is an opencast mine that produces manganese dioxide ore used by dry battery industry. This ore in the form of manganous oxide is used as micro-nutrient for cattle feed and fertilizers. MOIL fulfills about 50% of the total requirement of dioxide ore in India. At present, the annual production is around 1,093,363 tonnes which is expected to grow in the coming years. MOIL has set up Ferro Manganese Plant (10,000 TPY) and Electrolytic Manganese Dioxide (EMD) Plant (1500 TPY) as per its diversification plan for value addition to manganese ore. MOIL has also set up a Captive Power Plant and is further considering, expanding the capacity of ferro manganese plant and setting up a new Silico Manganese Plant by means of joint ventures entered into with Rashtriya Ispat Nigam Limited and Steel Authority of India Limited.

#### Name and address of the Applicant:

| Applicant                                                                                                                                                                                                                                                                                 | Nominated owner:-                                                                                                                                                                                                                                                                                    |
|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| M/s MOIL Limited.<br>(A Government of India Enterprise)<br>1A- MOIL Bhawan, Katol Road,<br>Nagpur-440 013<br><b>Telephone:</b> 0712-2590775<br><b>Fax:</b> 0712-2592073<br><b>E-mail:</b> moilind_ngp@sancharnet.in<br><b>Website:</b> <a href="http://www.moil.nic.m">www.moil.nic.m</a> | T.K.Pattnaik<br>Designation : Director (Commercial) & Addl<br>Charge Director(Production & Planning)<br>Add.- MOIL Limited, MOIL Bhawan,<br>1-A, Katol Road Nagpur- 440013 (MS)<br>Tel.No. : 0712-2592272<br>E-mail:- <a href="mailto:kishorchandraker66@gmail.com">kishorchandraker66@gmail.com</a> |

#### (iii) Brief Description of Nature of Project

Mining Plan and Progressive Mine Closure Plan over an area of 126.84 ha of ML in favour of MOIL in Village Gumgaon, Kodegaon and Tegai Tah: Saoner, Distt. Nagpur, Maharashtra State is submitted for approval under Rule 16 (1) of MCR 2016 and PMCP under Rule 23B of MCDR 1988 for 50 years.

#### (iv) Need for the project and its importance to the Country and or Region

Manganese is one of the major mineral deposits occurring in the Indian sub-continent. It has played a great role in development of civilization and industrialization. The

occurrence of Manganese ore in the proposed area is proved by the way of exploration and its production has important role in the steel industries.

The ore will be sold to Ferro/ Silico Industries. MOIL has its own specifications, which are derived as per buyer's demands & available quality. MOIL sales its ore to various Ferro/Silico producers as per demand. It is to be mentioned here that MOIL has set up its own lab, for chemical analysis the analytical results of the lab are expectable to all concerns

MOIL Limited, the leading producer of high grade Manganese ore and manufacturer of Electrolytic Manganese Dioxide and Manganese alloys in India, markets various grades and blends of ores to suits individual requirements of consumers, particularly for Steel, Ferro Manganese, Dry Battery, and Chemical Industries.

### 3.0 PROJECT DESCRIPTION

**(i) Type of Project including interlinked and interdependent projects, if any.**

This is a Manganese ore mining project, located at near village Gumgaon, Kodegaon and Tegai, Tahsil: Saoner, Distt; Nagpur- Maharashtra State. As per notification this project is more than 50.00 Hect. falls in 'A' category, as the area of lease is 126.84 Hectare. This is fresh environmental clearance case.

**(ii) Location detail:-**

The area falls with survey of India Toposheet No 55 K/15 and 55 O/3, the co-ordinates of all Boundary pillars are given on surface plan. The area is approachable by National & state highway from Nagpur. It is approximately 5 km from Patansongi on Patansaongi-Khapa road. This road is passing through the Gumgaon Mine of MOIL LTD. The area of 126.84 ha falls on west of Gumgaon Mine, which is contiguous. The area is undulating with gently slope due East. The general MRL is 310 and the highest MRL recorded is 331 MRL to the west. The hillock is standing in E-W direction, on North part of the area, made up of quartzite of Sitasongi formation. A Korardhari Nala, which is a tributary of river Kanhan flows along the East limit of the area. The granted ML area is private and revenue land. The private land is under cultivation. Two manganese old working pits and quarries are located in area. These pits are waterlogged.

### Co-ordinates of the Mining Lease area:-

| Pillar no. | Latitude       | Departure       | Pillar no. | Latitude        | Departure       |
|------------|----------------|-----------------|------------|-----------------|-----------------|
| BP1        | 21° 24' 12.34" | 78° 59' 7.56"   | BP64       | 21° 23' 47.15"  | 78° 58' 19.71"  |
| BP2        | 21° 24' 12.28" | 78° 59' 10.77"  | BP65       | 21° 23' 49.04"  | 78° 58' 14.95"  |
| BP3        | 21° 24' 4.27"  | 78° 59' 18.303" | BP66       | 21° 23' 52.70"  | 78° 58' 14.64"  |
| BP4        | 21° 24' 7.01"  | 78° 59' 20.41"  | BP67       | 21° 23' 56.93"  | 78° 58' 14.89"  |
| BP5        | 21° 24' 7.37"  | 78° 59' 19.61"  | BP68       | 21° 24' 0.95"   | 78° 58' 19.79"  |
| BP6        | 21° 24' 8.27"  | 78° 59' 19.68"  | BP69       | 21° 24' 5.48"   | 78° 58' 16.42"  |
| BP7        | 21° 24' 9.08"  | 78° 59' 19.48"  | BP70       | 21° 24' 8.83"   | 78° 58' 13.67"  |
| BP8        | 21° 24' 10.24" | 78° 59' 19.85"  | BP71       | 21° 24' 11.71"  | 78° 58' 25.40"  |
| BP9        | 21° 24' 10.31" | 78° 59' 20.09"  | BP72       | 21° 24' 8.51"   | 78° 58' 25.76"  |
| BP10       | 21° 24' 10.12" | 78° 59' 22.19"  | BP73       | 21° 24' 7.77"   | 78° 58' 31.79"  |
| BP11       | 21° 24' 10.76" | 78° 59' 22.78"  | BP74       | 21° 24' 5.19"   | 78° 58' 32.21 " |
| BP12       | 21° 24' 11.04" | 78° 59' 23.78"  | BP75       | 21° 24' 4.94"   | 78° 58' 33.51 " |
| BP13       | 21° 24' 11.55" | 78° 59' 24.80"  | BP76       | 21° 24' 1.89"   | 78° 58' 35.32"  |
| BP14       | 21° 24' 12.02" | 78° 59' 25.06"  | BP77       | 21° 24' 4.52"   | 78° 58' 35.59"  |
| BP15       | 21° 24' 11.88" | 78° 59' 29.29"  | BP78       | 21° 24' 6.34"   | 78° 58' 36.47"  |
| BP16       | 21° 24' 9.07"  | 78° 59' 30.59"  | BP79       | 21° 24' 7.13"   | 78° 58' 36.09"  |
| BP17       | 21° 24' 8.85"  | 78° 59' 32.90"  | BP80       | 21° 24' 7.77"   | 78° 58' 38.27"  |
| BP18       | 21° 24' 6.26"  | 78° 59' 33.15"  | BP81       | 21° 24' 6.69"   | 78° 58' 38.40"  |
| BP19       | 21° 24' 5.37"  | 78° 59' 34.87"  | BP82       | 21° 24' 2.34"   | 78° 58' 37.29"  |
| BP20       | 21° 24' 2.82"  | 78° 59' 35.55"  | BP83       | 21° 23' 59.49"  | 78° 58' 39.87"  |
| BP21       | 21° 24' 1.71"  | 78° 59' 36.90"  | BP84       | 21° 23' 58.77"  | 78° 58' 46.48"  |
| BP22       | 21° 24' 0.31"  | 78° 59' 37.03"  | BP85       | 21° 23' 56.46"  | 78° 58' 52.54"  |
| BP23       | 21° 23' 58.74" | 78° 59' 39.66"  | BP86       | 21° 23' 58.62"  | 78° 58' 56.14"  |
| BP24       | 21° 23' 55.72" | 78° 59' 40.99"  | BP87       | 21° 23' 58.61 " | 78° 58' 59.84"  |
| BP25       | 21° 23' 55.36" | 78° 59' 40.04"  | BP88       | 21° 23' 59.45"  | 78° 59' 1.37"   |
| BP26       | 21° 23' 55.98" | 78° 59' 38.18"  | BP89       | 21° 24' 0.04"   | 78° 59' 3.06"   |
| BP27       | 21° 23' 54.50" | 78° 59' 38.10"  | BP90       | 21° 24' 7.04"   | 78° 59' 4.38"   |
| BP28       | 21° 23' 54.55" | 78° 59' 36.06"  | BP91       | 21° 24' 6.39"   | 78° 59' 7.40"   |
| BP29       | 21° 23' 52.31" | 78° 59' 34.20"  | BP92       | 21° 24' 16.23"  | 78° 58' 36.59"  |
| BP30       | 21° 23' 51.64" | 78° 59' 34.23"  | BP93       | 21° 24' 15.66"  | 78° 58' 31.94"  |
| BP31       | 21° 23' 51.79" | 78° 59' 32.05"  | BP94       | 21° 24' 13.50"  | 78° 58' 25.55"  |
| BP32       | 21° 23' 53.32" | 78° 59' 31.55"  | BP95       | 21° 24' 12.56"  | 78° 58' 12.83"  |

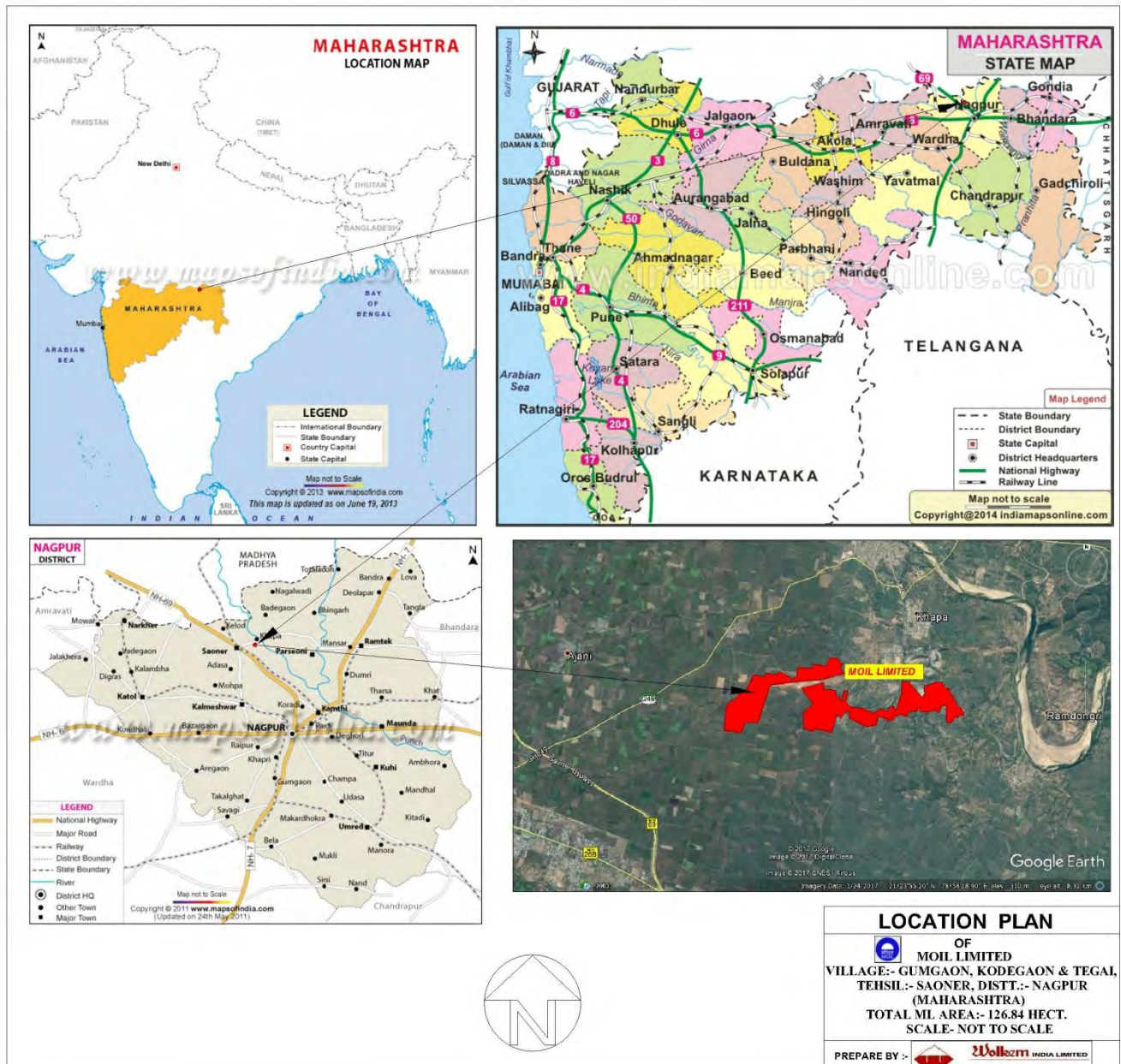
|      |                |                |       |                |                |
|------|----------------|----------------|-------|----------------|----------------|
| BP33 | 21° 23' 52.94" | 78° 59' 31.28" | BP96  | 21° 24' 10.48" | 78° 57' 56.92" |
| BP34 | 21° 24' 0.38"  | 78° 59' 31.48" | BP97  | 21° 24' 1.66"  | 78° 57' 53.57" |
| BP35 | 21° 24' 0.41"  | 78° 59' 28.83" | BP98  | 21° 23' 51.94" | 78° 57' 50.26" |
| BP36 | 21° 23' 56.61" | 78° 59' 27.11" | BP99  | 21° 23' 51.94" | 78° 57' 43.15" |
| BP37 | 21° 23' 57.97" | 78° 59' 26.07" | BP100 | 21° 23' 46.82" | 78° 57' 41.31" |
| BP38 | 21° 23' 57.06" | 78° 59' 19.49" | BP101 | 21° 23' 47.84" | 78° 57' 32.02" |
| BP39 | 21° 23' 57.58" | 78° 59' 17.81" | BP102 | 21° 24' 0.18"  | 78° 57' 33.10" |
| BP40 | 21° 23' 59.24" | 78° 59' 16.82" | BP103 | 21° 24' 5.03"  | 78° 57' 39.70" |
| BP41 | 21° 23' 57.48" | 78° 59' 15.35" | BP104 | 21° 24' 7.42"  | 78° 57' 40.64" |
| BP42 | 21° 23' 57.47" | 78° 59' 13.89" | BP105 | 21° 24' 8.25"  | 78° 57' 43.96" |
| BP43 | 21° 23' 56.74" | 78° 59' 11.36" | BP106 | 21° 24' 15.67" | 78° 57' 47.64" |
| BP44 | 21° 23' 58.15" | 78° 59' 10.38" | BP107 | 21° 24' 14.38" | 78° 57' 54.08" |
| BP45 | 21° 23' 54.74" | 78° 59' 10.53" | BP108 | 21° 24' 16.47" | 78° 57' 54.47" |
| BP46 | 21° 23' 55.92" | 78° 59' 6.93"  | BP109 | 21° 24' 15.65" | 78° 57' 59.17" |
| BP47 | 21° 23' 55.62" | 78° 59' 5.89"  | BP110 | 21° 24' 16.87" | 78° 58' 2.33"  |
| BP48 | 21° 23' 54.74" | 78° 59' 3.57"  | BP111 | 21° 24' 16.84" | 78° 58' 3.06"  |
| BP49 | 21° 23' 53.34" | 78° 59' 3.16"  | BP112 | 21° 24' 16.11" | 78° 58' 4.26"  |
| BP50 | 21° 23' 52.86" | 78° 59' 1.12"  | BP113 | 21° 24' 15.45" | 78° 58' 4.99"  |
| BP51 | 21° 23' 55.27" | 78° 58' 54.74" | BP114 | 21° 24' 14.54" | 78° 58' 6.61"  |
| BP52 | 21° 23' 50.39" | 78° 58' 53.63" | BP115 | 21° 24' 13.14" | 78° 58' 8.61"  |
| BP53 | 21° 23' 50.54" | 78° 58' 53.13" | BP116 | 21° 24' 15.13" | 78° 58' 11.21" |
| BP54 | 21° 23' 49.67" | 78° 58' 48.62" | BP117 | 21° 24' 19.44" | 78° 58' 11.48" |
| BP55 | 21° 23' 53.75" | 78° 58' 39.72" | BP118 | 21° 24' 18.95" | 78° 58' 19.37" |
| BP56 | 21° 23' 54.60" | 78° 58' 39.17" | BP119 | 21° 24' 20.77" | 78° 58' 19.56" |
| BP57 | 21° 23' 56.87" | 78° 58' 39.56" | BP120 | 21° 24' 20.33" | 78° 58' 25.15" |
| BP58 | 21° 23' 57.22" | 78° 58' 38.94" | BP121 | 21° 24' 19.75" | 78° 58' 25.61" |
| BP59 | 21° 23' 56.86" | 78° 58' 37.02" | BP122 | 21° 24' 19.76" | 78° 58' 26.41" |
| BP60 | 21° 23' 54.41" | 78° 58' 34.77" | BP123 | 21° 24' 24.33" | 78° 58' 28.77" |
| BP61 | 21° 23' 55.96" | 78° 58' 32.38" | BP124 | 21° 24' 23.98" | 78° 58' 32.88" |
| BP62 | 21° 23' 46.88" | 78° 58' 29.82" | BP125 | 21° 24' 20.90" | 78° 58' 35.45" |
| BP63 | 21° 23' 48.87" | 78° 58' 20.45" | BP126 | 21° 24' 19.09" | 78° 58' 34.36" |

**Land Detail:-**

| Village                                  | Khasra No.                                       | Lease Area Ha. | Ownership/ Occupancy                               |
|------------------------------------------|--------------------------------------------------|----------------|----------------------------------------------------|
| Village - Gumgaon,<br>Kodegaon and Tegai | As per list enclosed at<br><b>Annexure – III</b> | 126.46 Ha.     | Private land – 113.82ha.<br>Govt. Land – 12.64 Ha. |

**Schedule of the Area:-**

| District                           | Village            | Khasra No.   | Area in Ha. | Status    | Khasra No. | Area in Ha.      | Status   |
|------------------------------------|--------------------|--------------|-------------|-----------|------------|------------------|----------|
| Distt:<br>Nagpur<br>Teh:<br>Saoner | Gumgaon            | 8            | 4.46        | Pvt.land  | 176        | 1.48             | Pvt.land |
|                                    |                    | 157          | 1.30        | Pvt.land  | 177        | 0.86             | Pvt.land |
|                                    |                    | 158          | 1.60        | Pvt.land  | 178        | 1.25             | Pvt.land |
|                                    |                    | 166/1        | 1.21        | Pvt.land  | 179        | 1.60             | Pvt.land |
|                                    |                    | 166/2        | 2.03        | Pvt.land  | 188        | 0.05             | Pvt.land |
|                                    |                    | 167          | 0.38        | Pvt.land  | 189        | 1.27             | Pvt.land |
|                                    |                    | 168          | 1.29        | Pvt.land  | 190        | 1.14             | Pvt.land |
|                                    |                    | 169          | 0.80        | Pvt.land  | p.o. 192   | 0.97             | Pvt.land |
|                                    |                    | 170          | 0.85        | Pvt.land  | 193        |                  | Pvt.land |
|                                    |                    | 171          | 0.91        | Pvt.land  | 194        | 0.68             | Pvt.land |
|                                    |                    | 172          | 3.91        | Pvt.land  |            |                  |          |
|                                    |                    | 173          | 2.31        | Pvt.land  |            |                  |          |
|                                    |                    | <b>Total</b> |             |           |            | <b>30.35 Ha.</b> |          |
|                                    | Tegai              | 1            | 2.22        | Pvt.land  | 68         | 0.96             | Pvt.land |
|                                    |                    | 2            | 0.08        | Pvt.land  | 69         | 0.60             | Pvt.land |
|                                    |                    | 3            | 1.78        | Pvt.land  | 70         | 0.50             | Pvt.land |
|                                    |                    | 4            | 0.95        | Pvt.land  | 71         | 1.10             | Pvt.land |
|                                    |                    | 10           | 1.94        | Pvt.land  | 72/1       | 2.06             | Pvt.land |
|                                    |                    | 12           | 1.48        | Pvt.land  | 72/2       | 1.16             | Pvt.land |
|                                    |                    | 13           | 1.04        | Pvt.land  | 72/3       | 0.90             | Pvt.land |
|                                    |                    | 14           | 1.58        | Pvt.land  | 73/1       | 0.72             | Pvt.land |
|                                    |                    | 15a          | 0.28        | Pvt.land  | 73/2       | 0.82             | Pvt.land |
|                                    |                    | 15b          | 3.02        | Pvt.land  | 73/3       | 0.74             | Pvt.land |
|                                    |                    | 15 k/1       | 1.00        | Pvt.land  | 142/1      | 0.81             | Pvt.land |
|                                    |                    | 15k/2        | 1.02        | Pvt.land  | 142/2      | 0.83             | Pvt.land |
|                                    |                    | 16           | 0.56        | Pvt.land  | 142/3      | 1.33             | Pvt.land |
|                                    |                    | 17           | 1.52        | Pvt.land  | 142/4      | 1.58             | Pvt.land |
|                                    |                    | 63           | 1.00        | Pvt.land  | 143        | 1.21             | Pvt.land |
|                                    |                    | 64           | 1.18        | Pvt.land  | 144        | 2.48             | Pvt.land |
|                                    |                    | 65           | 1.10        | Pvt.land  | 145        | 0.56             | Pvt.land |
|                                    |                    | 66           | 1.00        | Pvt.land  | 146        | 0.52             | Pvt.land |
|                                    |                    | 67           | 1.20        | Pvt.land  | 148        | 3.16             | Pvt.land |
|                                    |                    | <b>Total</b> |             |           |            | <b>45.99 Ha.</b> |          |
|                                    | Kodegaon           | 94           | 3.51        | Pvt.land  | 136        | 0.49             | Pvt.land |
|                                    |                    | p.o 95       | 1.26        | Pvt.land  | 137        | 1.54             | Pvt.land |
|                                    |                    | 96           | 5.76        | Govt.land | 138-278    | 1.60             | Pvt.land |
|                                    |                    | 97           | 0.96        | Pvt.land  | 139        | 1.10             | Pvt.land |
|                                    |                    | 122-123/1    | 0.97        | Pvt.land  | 140        | 5.59             | Pvt.land |
|                                    |                    | 122-123/2    | 1.21        | Pvt.land  | 141        | 4.17             | Pvt.land |
|                                    |                    | p.o. 126     | 1.88        | Govt.land | 142        | 2.06             | Pvt.land |
|                                    |                    | 131          | 2.13        | Pvt.land  | 143/1      | 4.10             | Pvt.land |
|                                    |                    | 132          | 3.67        | Pvt.land  | 143/2      | 0.80             | Pvt.land |
|                                    |                    | 133          | 2.28        | Pvt.land  |            |                  |          |
|                                    |                    | 134          | 5.00        | Govt.land |            |                  |          |
|                                    |                    | 135          | 0.42        | Pvt.land  |            |                  |          |
|                                    |                    | <b>Total</b> |             |           |            | <b>50.50 Ha.</b> |          |
|                                    | <b>Grand Total</b> |              |             |           |            | <b>126.84 Ha</b> |          |



(iii) Details of alternate sites considered and the basis of selecting the proposed site, particularly the environmental considerations gone into should be highlighted.  
No alternative site has been considered.

(iv) Size and Magnitude of Operation

The proposed mining Lease Area is 126.84 Ha. With production capacity is proposed annually up to 1,02,000 T manganese ore of ROM when the mine will be fully developed.

**(v) Project description with process details (a schematic diagram/flow chart showing the project layout, components of the project etc. should be given)**

**Geology:-**

**• Regional Geology:-**

The Precambrian (Mesoproterozoic to Neoproterozoic) meta-sedimentary sequence of Sausar Group is well known for some of the largest manganese ore deposits in Central India. The Sausar Fold Belt (SFB) is an important constituent of the Central Indian Tectonic Zone (CITZ) - a crustal-scale Precambrian mobile belt running E-W through the Indian Peninsular Shield. It constitutes the southern boundary of CITZ and is bounded by Central Indian Suture zone (CIS) on its south. It is 300 km long and 70 km wide, curvilinear (convex towards south) in shape, ESE-WNW to E-W to ENE-WSW trending orogenic belt bounded by two cratonic blocks, the Bundelkhand Craton (BKC) in the north and Bastar Craton (BC) in the south (Roy et al., 2006). The central portion comprises of a stable platform sequence of manganese bearing pelite-arenite-carbonate rocks along with Tirodi Biotite Gneiss (TBG) and intrusive granitoids. The Sausar supracrustal rocks have undergone polyphase deformation (Bhowmik et al., 1999) and the dominant E-W structural trend in SFB is a result of the combined effect of multiple phases of deformations (Chattopadhyay et al., 2003). Deformation produced low angle thrusting, tight-to-isoclinal, recumbent-to-reclined folds which led to the tectonic interleaving of basement and supracrustal rocks which resulted in co-folding of the basement gneisses with the supracrustal rocks at many places. The metamorphic grade in the Sausar supracrustal rocks vary from green schist to upper amphibolite facies with gradual increase in the grade of metamorphism from east-southeast to northwest (Bhowmik et al., 1999). However, the gneissic rocks adjacent to the Sausar Group in the north and south contain enclaves of pelitic and basic granulites, overprinted by a retrograde amphibolites facies fabric (Bhowmik et al., 1999). The Bhandara-Balaghat Granulite (BBG) domain to the south includes mafic granulite, charnokite, enderbite, meta-ultrabasites, cordierite granulite and kyanite quartzite. Whereas Ramakona- Katangi Granulite belt (RKG) to the north consists mainly of cordierite gneiss, mafic granulite, felsic migmatitic gneiss and foliated gneisses.

Recent geo-chronological study has indicated that the SFB experienced a major tectonothermal event around 850-950 Ma (Roy et al., 2006). The above tectonothermal event imprinted an amphibolite facies fabric over 1100 Ma granulite grade foliations in the high grade quartzofeldspathic gneisses of the TBG suite. It is suggested that these tectonic domains in the Sausar fold belt are the result of early subduction in the southern part of CITZ followed by continent-continent collision from south to north around Grenvillian time (Roy and Prasad, 2003; Bhowmik, 2006; Bhowmik, et al. 2012).

The stratigraphic succession of the Sausar Group is shown in table-1 (Chattopadhyay, et al., 2003). The Mansar formation which contains significant amount of manganese ore horizon associated with gondite rocks is rich in mica schist and muscovite-biotite-schist. The underlying Lohangi formation is composed of Dolomite marble with lenses of manganese ore and gondite rocks and rests over the Archean TBG basement. The overlying Chorbaoli formation consists of quartzite, quartz-schist, quartz muscovite schist and Bichua formation occur as dolomitic marbles with calc-silicate and schistose bands and sillimanite bearing quartz biotite-granulite. All the above formations are intruded by pegmatite, quartz vein and intrusive granitoids. One of the most characteristic features of this metamorphic belt is the absence of mafic rocks. Presence of a large granite body intruding the schistose rocks immediately underlying the quartzite-pelite sequence in the Mansar area indicates that the granite emplacement was syntectonic (Chattopadhyay, et al., 2003).

The ore bodies invariably occupy the Mansar formation ( in the middle part , at the top and bottom of the Mansar formation in contact with overlying and underlying rocks) (Straczek et al., 1956). In all cases, the manganese silicate rocks (gondites) are uniformly interbanded with manganese oxide ore bodies (braunite, bixbyite, hollandite, hausmannite, jacobsonite, manganite, pyrolusite, cryptomelane), and are often co-folded in different scales. Gondite is a metamorphosed manganiferous sedimentary rock consisting of quartz and spessartite as essential minerals. However, braunite, rhodonite and other manganese silicates are also very common (Straczek et al., 1956). The individual manganese silicate bands vary in thickness from a few cm to several meters. The mineral assemblages in gondites of different metamorphic zones suggest that they are formed due to regional metamorphism of mixed manganiferous sediments that were originally laid down in the oxidizing condition (Roy and Mitra, 1964). After the careful assessment of mineralogy, texture and paragenesis of the manganese ore of the study area, Jawed and Siddiquie (2014) have concluded that the manganese ores of the Nagpur area are formed by the multiple processes i.e. metamorphism and supergene enrichment.

| Formations  |                            |                       | Litho Units                                                                                                             |
|-------------|----------------------------|-----------------------|-------------------------------------------------------------------------------------------------------------------------|
| Recent      |                            |                       | Alluvium.<br>Unconformity                                                                                               |
| Proterozoic | S<br>a<br>u<br>s<br>a<br>r | Intrusive             | Pegmatite, granites and vein quartz.                                                                                    |
|             |                            | Bichua formation      | Dolomite marble, calc silicate rocks, sillimanite bearing quartz-bio-granulites, biotite-muscovite-schist,              |
|             | G<br>r<br>o<br>u<br>p      | Chorbaoli formation   | Garnet-staurolite-quartz-muscovite schist                                                                               |
|             |                            | Mansar formation      | Manganese ore I and gondite muscovite schist, Manganese ore II and schist, Manganese ore III with gondite               |
|             |                            | Lohangi formation     | Dolomite marbles with lenses of manganese ore, calc-silicate rocks, calc-granulites, quartz-bio-granulites and gneisses |
| Archaean    |                            | Tirodi biotite gneiss | Biotite gneiss, migmatite, tonalite gneiss, cordierite gneiss, amphibolites etc.                                        |

### Local Geology:

The manganese ore horizon of Gumgaon deposit belongs to Munsar Formation of Saucer Group. The deposit is being mined out by underground method of mining by MOIL which is adjoining to applied area of 126.84 Ha. As the underground mine is fully developed and work has been carried since beginning of 20th Century the structure has already been established. This zone is displayed in anticline form with a strike east west and dip due south with moderate to high angle. This fold also plunges due west with low angle. The deposit is faulted with a vertical fault trending in north east – south west direction, dislocating the eastern block to south. This deposit has been explored to the depth of –1200'L with 29 boreholes. The avg. thickness available for mining within the area ranges from 3 m to 43 m.

### Structure:

In Gumgaon mine the ore body and the country rock have a similar trend N730E to S730 W and dip 550 to 750 due South. The ore bed shows westerly plunge and amount varies from 80 to 120 as shown in the cross sections.

### Folds:

Structurally the ore body is divided into two ore bands which form the two limbs of an isoclinal anticline. Northern limb contains the main bed which is main ore producing limb; the southern limb contains south bed which is exhausted. An auxiliary ore band occurring on surface may be dragged portion of the north bed. The Chorbaoli Quartzites occurring on the north west of quarry and a small exposure on PWD road in the west form a core of syncline which lies north of this anticline. Old south bed contained southern limb of anticline and was worked earlier. The ore body is pinched out in dip as well as in strike direction. This phenomenon is proved by boreholes namely 18, 20 & 21 winzes and drives. The major fault trending NNE-SSW has affected the south limb and has terminated its westward lateral extension.

### Faults:

There is a NNE-SSW fault passing through the main bed very near to its eastern end. The net down throw of the ore body is about 90 mtr to the west. The ore body is dragged along the main fault where it is highly crushed. Besides the main fault there are large numbers of small slips having a throw of 0.50 mtrs.

### Joints:

There are various sets of joints out of which three sets are of utmost importance as their behavior controls the mining operation in underground. One set is striking N-S and dipping with various angles to north, the third one is also N-S trending and vertical. Whenever vertical joints occur in parallel fashion there is combination with east –west trending joints which create a weak zone and there is always possibility of formation of jointed blocks which is liable to fall under its own weight. The other important set of east west trending joints with northerly dips

merge with weak bedding planes of the strata, it gives rise to wedge shaped block formation which due to its emerging base is dangerous.

**Mineralogy of ore:** based on examination of polished section under microscope the following minerals assemblage are most commonly observed.

1. Braunite- Hausmanite- Jacobsite.
2. Braunite-Cryptomelane-Pyrolusite
3. Braunite-Psilomelane

**Chemistry of ore:** Mainly primary Manganese grades and subordinate quantity of secondary Manganese oxide ores are being present. The marketability is controlled by the chemistry of ore, the chemical variations in various ore types are controlled by associated amounts of silica minerals. Other impurities which having bearing on its commercial utilization are iron and phosphorus contents. The chemical analysis gives the following range of constitutes

|      |                   |       |                 |
|------|-------------------|-------|-----------------|
| Mn   | 25.00% to 50.00%  | CaO   | 1.00% to 5.50 % |
| Fe   | 5.00% to 9.00 %   | MgO   | 2.00% to 4.00 % |
| Sio2 | 7.00 % to 22.00 % | Al2O3 | 1.50 % to 4.00% |
| P    | 0.15 % to 0.30 %  |       |                 |

From the above analytical result it is clear that in this area manganese is from low to high grade with characteristically medium range of phosphorus impurities. That makes these deposits important from commercial point of view.

**Metamorphism:** Meta sedimentary rocks of the area represented by vitreous and Micaceous quartzite, Muscovite and Biotite Schists, Biotite Gneiss and Granulites. Geniss and Granulites are mainly Feldspthised products of Schists. Texturally different rock types, ore and ore minerals are characteristically uniformly coarse grained. Thus the enclosing lithology of the Manganese ore is the pelitic schists and quartzites. Metamorphic mineralogy of the whole assemblage is as below.

Oxide: Braunite, Jacobsite, Hausmanite/Cryptomelane, Psilomelane and Pyrolusite.

**Silicates:** Biotite, Muscovite, Quartz, Sillimanite, and Garnet.

The above mineral assemblage and coarse texture of rocks and ore is indicative of Sillimanite Garnet –Muscovite sub facies of metamorphism. Possible pre metamorphic character of manganiferous sediments may be explained to have been sharply inter bedded manganese oxide (with minor Fe & Si) and admixed manganese oxide, clay silica sediments. Thus the pelitic sediments with the above composition have undergone a medium grade regional metamorphism which has resulted into formation of present lithology.

## Reserve

**Proved Mineral Reserves (121):- 7,84,856 Tonnes**

## Resources:-

**Prefeasibility Mineral Resource (222): 69,230 Tonnes**

Between section 4 & section 5 the lease area was not extensively explored hence the quantum of this area is calculated under Prefeasibility Mineral Resource (222).

**Inferred Mineral Resource (333):- 2,80,000 Tonnes.**

The ore zone below 400 m is consider under indicated mineral Resource (333), as the ore body reveals the continuity. Hence the quantum computed is placed as Inferred Mineral Resource (333) of UNFC.

During the span of 5 years the exploration by core drilling will be conducted over balance area, for change in UNFC category of Geological Axis to detail exploration in 5 years time. On obtaining the Geological data, the feasibility of the project will be decided and according the Mining shall be continued.

## Method of Mining:-

### A. OPEN CAST MINING:

There is no proposal in this area for opencast mining.

### B. UNDERGROUND MINING:

At present, there are no mining activities in this ML area of 126.84 ha. The mode of entry in to this area in the underground will be through the proposed 2nd vertical shaft of 5.5 m dia which will be sunk at Ch. 5200 in lease 85.90 ha of Gumgoan mine, MOIL LTD. Vertical shaft will serve 7 levels viz -300'L, -400'L, -500'L, -600'L, -700', -800'L and 900'L. This

shaft will be sunk from 306 GSL to -18 underground RL for a depth of 324 m. Presently, mode on entry to 85.90 ha area through the vertical shaft which is located at Ch. 23 in lease 85.90 ha and will serve up to -400'L. Shaft inset has been done up to -400'L. The depth of the present vertical shaft is 169 m below the surface.

In the ML area of 126.84 ha, mining activity will be commenced after execution of lease. The proposed vertical will be sunk from the Year 2017-18 and will be commissioned in the year 2021-22 with infrastructural development at -500'L. Initially development of haulage road from 2nd Vertical shaft will be carried in lease 85.90 ha in the year 2021-22 for 550 mtr to enter in this lease area. Thereafter permission will be taken to develop drive below the nallah by leaving insitu barrier of 204 m below surface from the floor of the nallah bed. No mining activities will be carried out under the nallah except drift development. All the

statutory provisions will be adopted as per the mines Act 1952 and Regulation MMR, 1961

After entry in the lease area Haulage development will be carried out in 500'L and 600'L and winzes will be dome from -500'L to - 600;L . Haulage drive will be developed by leaving 7.5 m pillar for lease boundary and will enter in the other block of the lease area in the year 2023-24.

### C. Mode of entry:

Opening of lower levels for entire strike length is a constraint due to the limitations of lease hold area. Owing to the nature and behavior of ore zone with depth, the lower levels will be only partially within the lease. Thus partial extraction is only possible. In absence of availability of additional lease area on foot wall side, mining activities will be restricted to limited extent and depth.

The underground levels were opened by development of haulage road and winzes. To connect the underground levels, the winzes are driven at regular interval of 60 mtrs, thus forming a block of 60 m x 60 m. size.

The 2nd Vertical Shaft will be sunk at Ch. 5200 from surface of lease and will connect the underground levels from -300'L. Details of the present vertical shaft located at Ch. 2300 at lease 85.90 ha and Proposed 2nd vertical shaft is also located at Ch. 5200 in this lease.

### Details of Vertical Shaft at Gumgaon Mine:

The finished diameter of the vertical shaft is 4.5 m. It is located at Ch. 23 in lease over an area of 85.90 ha. The shaft has been sunk up to 169 mtrs from the surface. Hoisting capacity of the present shaft is 1,20,000 Tonnes ROM per year.

| Moil Limited, Present Vertical Shaft at Gumgaon Mine |                                    |                                      |
|------------------------------------------------------|------------------------------------|--------------------------------------|
| S.N.                                                 | Description of work                | Details                              |
| 1                                                    | Shape of shaft                     | Circular shaft                       |
| 2                                                    | Size of shaft                      | 4.5 mtrs. dia.                       |
| 3                                                    | Depth of proposed sinking/deepenig | 169 mtrs                             |
| 4                                                    | Do. of levels                      | 4 levels                             |
| 5                                                    | Dinding                            | Cage winding with tub                |
| 6                                                    | Tub loading arrangement            | at each level                        |
| 7                                                    | Hoisting capacity                  | 1,20,000 ROM/TPA                     |
| 8                                                    | Total depth of shaft from surface  | 169 mtrs                             |
| 9                                                    | Pit top arrangement                | O.H. bunker of 100t with tub tippler |
| 10                                                   | Pit bottom arrangement             | Suitable for movement of tubs        |
| 11                                                   | Commissioned on                    | 2012                                 |

## Details of the Proposed 2nd Vertical Shaft of Gumgaon Mine

### Salient Features

The finished diameter of the proposed shaft will be 5.5 m. It is proposed to sink this shaft at Ch.52, as per details given below, for producing 1,80,000 Tonnes (shaft capacity) ROM. Location at Ch. 52 in lease over an area of 85.90 ha

| S.N. | Description Of Work               | Details                              |
|------|-----------------------------------|--------------------------------------|
| 1    | Shape Of Shaft                    | Circular Shaft                       |
| 2    | Size Of Shaft                     | 5.5 Mtrs. Dia.                       |
| 3    | Depth Of Proposed Sinking         | 324 Mtrs                             |
| 4    | No. Of Levels                     | 07 Levels                            |
| 5    | Winding                           | Cage Winding With Tub                |
| 6    | Tub Loading Arrangement           | At Each Level                        |
| 7    | Hoisting Capacity                 | 180000 ROM/TPA                       |
| 8    | Total Depth Of Shaft From Surface | 324 Mtrs                             |
| 9    | Pit Top Arrangement               | O.H. Bunker Of 100t With Tub Tippler |
| 10   | Pit Bottom Arrangement            | Suitable For Movement Of Tubs        |
| 11   | Period Of Completion              | 4 Years                              |
| 12   | Sinking                           | 324 Mtr.                             |
| 13   | Proposed Commencement Of Work     | 2017-18                              |
| 14   | Proposed Completion Of Work       | 2021-22                              |

### The Proposed Production Programme:-

Year wise development and production for 1<sup>th</sup> to 5<sup>th</sup> Year

#### Year Wise Development

| Year | Horizontal Dev             |                              |                         |                     |                         | Vertical Dev                |
|------|----------------------------|------------------------------|-------------------------|---------------------|-------------------------|-----------------------------|
|      | HR - RMT<br>3.0m X<br>3.5m | X-cut -RMT<br>3.0m X<br>3.5m | Total<br>RMT<br>in Rock | OD-RMT<br>2.1mx1.8m | Horizontal<br>Dev Total | Winzes-<br>RMT<br>1.8mx1.8m |
| 1    | 0                          | 0                            | 0                       | 0                   | 0                       | 0                           |
| 2    | 0                          | 0                            | 0                       | 0                   | 0                       | 0                           |
| 3    | 0                          | 0                            | 0                       | 0                   | 0                       | 0                           |
| 4    | 0                          | 0                            | 0                       | 0                   | 0                       | 0                           |
| 5    | 0                          | 0                            | 0                       | 0                   | 0                       | 0                           |

|       |   |   |   |   |   |   |
|-------|---|---|---|---|---|---|
| Total | 0 | 0 | 0 | 0 | 0 | 0 |
|-------|---|---|---|---|---|---|

### ROM Generation during Development

| Year  | OD-RMT<br>2.1mx1.8m | ROM in<br>Tonnes | Winzes-RMT<br>1.8mx1.8m | ROM<br>in Tonnes | Total<br>ROM in T |
|-------|---------------------|------------------|-------------------------|------------------|-------------------|
| 1     | 0                   | 0                | 0                       | 0                | 0                 |
| 2     | 0                   | 0                | 0                       | 0                | 0                 |
| 3     | 0                   | 0                | 0                       | 0                | 0                 |
| 4     | 0                   | 0                | 0                       | 0                | 0                 |
| 5     | 0                   | 0                | 0                       | 0                | 0                 |
| Total | 0                   | 0                | 0                       | 0                | 0                 |

### Waste Generation During Development in m<sup>3</sup>

| Year  | HR - RMT<br>3.0m X 3.5m | X-cut -RMT<br>3.0m X 3.5m | Total RMT<br>in Rock | Generation<br>of Waste in m <sup>3</sup> |
|-------|-------------------------|---------------------------|----------------------|------------------------------------------|
| 1     | 0                       | 0                         | 0                    | 0                                        |
| 2     | 0                       | 0                         | 0                    | 0                                        |
| 3     | 0                       | 0                         | 0                    | 0                                        |
| 4     | 0                       | 0                         | 0                    | 0                                        |
| 5     | 0                       | 0                         | 0                    | 0                                        |
| Total | 0                       | 0                         | 0                    | 0                                        |

### Yearly Proposed Generation of ROM and Clean Ore

| Year  | No of Stoles | ROM in m <sup>3</sup> | ROM in T | Clean Ore in T |
|-------|--------------|-----------------------|----------|----------------|
| 1     | 0            | 0                     | 0        | 0              |
| 2     | 0            | 0                     | 0        | 0              |
| 3     | 0            | 0                     | 0        | 0              |
| 4     | 0            | 0                     | 0        | 0              |
| 5     | 0            | 0                     | 0        | 0              |
| Total | 0            | 0                     | 0        | 0              |

### Proposed Development, ROM, Clean Ore and Reject Generation for 1<sup>st</sup> to 5<sup>th</sup> Year

| Year                  | 1 | 2 | 3 | 4 | 5 | Total |
|-----------------------|---|---|---|---|---|-------|
| Dev in m <sup>3</sup> | 0 | 0 | 0 | 0 | 0 | 0     |
| ROM in T              | 0 | 0 | 0 | 0 | 0 | 0     |
| Clean Ore in T        | 0 | 0 | 0 | 0 | 0 | 0     |
| Rejects in T          | 0 | 0 | 0 | 0 | 0 | 0     |

### Proposed Requirement of Sand from 1<sup>st</sup> to 5<sup>th</sup> year

| Year    | 1 | 2 | 3 | 4 | 5 | Total |
|---------|---|---|---|---|---|-------|
| Sand in | 0 | 0 | 0 | 0 | 0 | 0     |

### Conceptual Mine Plan & Life of Mine

#### Life of Mine

|                        |        |
|------------------------|--------|
| Reserves/Resouces in T | 854086 |
| Mineable in T          | 640464 |
| Yearly Av Prod of ROM  | 17030  |
| Life of Mine           | 37.60  |
| Say 38 Years           |        |

### As per Mining Plan

#### D. Machinery:-

- Lathe Machine : 01 No.
- Drill Machine : 01 No.
- Power Saw Machine : 01 No.
- Welding Machine : 01 No

All above machines are in good condition and properly lubricated. Provision of suitable safety guards and cutting lubrication has been made to carry out safe operation.

### Details of the pumps:

| Sr. No. | Description               | Quantity | Location           | H.P. | Capacity | Head     |
|---------|---------------------------|----------|--------------------|------|----------|----------|
| 1       | DSM Pump                  | 02       | +13'L              | 75   | 1800 LPM | 90 Mtr.  |
| 2       | Maxflow make Pump         | 01       | +13'L              | 167  | 1500LPM  | 140 Mtr  |
| 3       | Maxflow make Pump         | 02       | -100'L             | 125  | 2250LPM  | 169Mtr.  |
| 4       | Bareja make Pump          | 01       | -100'L             | 90   | 1800LPM  | 150Mtr.  |
| 5       | Maxflow make Pump         | 01       | -200'L             | 25   | 950LPM   | 90Mtr.   |
| 6       | Kirloskar make Pump       | 01       | -200'L             | 200  | 3000LPM  | 300Mtr   |
| 7       | Maxflow make Pump         | 01       | -200'L             | 200  | 3000PM   | 300 Mtr  |
| 8       | Maxflow make Pump         | 01       | -200'L             | 220  | 3000PM   | 300 Mtr  |
| 9       | Mather & Plate make pump  | 01       | 100'L              | 100  | 2000 LPM | 100 MTR. |
| 10      | MBH make submersible pump | 02       | - 400 L pit bottom | 100  | 4000 LPM | 65 Mtr.  |
| 11      | MBH make submersible pump | 02       | - 400 L pit bottom | 50   | 1800 LPM | 80 Mtr.  |

### Locomotive:

At Gumgaon Mine ore cleaning floor (Surface) is being done by diesel locomotive. One locomotive is quite old and imported i.e. Simplex loco English make. The loco however is re powered with suitable Kirloskar make Diesel Engine and is still in service. Two locomotives are available and put in service for surface ore transportation. These are operated in both shifts. Maintenance of locomotive is being done as per schedule.

Ovis Diesel locomotive - 01

Simplex Diesel locomotive - 01

Suri & Nair Diesel locomotive - 01

#### D.G. Set:

Gumgaon mine is having 1500 KVA D.G. set for power supply of emergency load and winder in the event of utility supply failure. This D.G. set is installed at Compressor house.

#### COMPRESSORS:

For underground drilling we have following electric operated compressors which are stationed at surface.

#### Details of Compressor

| Sr. no. | Make                                                  | Quantity | H.P. | Capacity |
|---------|-------------------------------------------------------|----------|------|----------|
| 01      | Electric operated<br>I.R. make, Compressor with VFD   | 01       | 250  | 1000 CFM |
| 02      | Electric operated<br>I.R. make, Compressor            | 01       | 250  | 1000 CFM |
| 03      | Electric operated<br>Godrei Make, Compressor with VFD | 01       | 200  | 1000 CFM |

#### E. Extent of Mechanization:

##### System of Drilling and blasting:

Drilling in the drifts for haulage road, x-cut and ore drive will be done by pneumatically operated jack hammer of 33 mm dia of the length of 1.5 m. Generally Wedge cut and parallel cut is being carried out at Gumgaon Mine. For Mechanized development drift dimensions will be change 3.0 m x 3.5 m for introduction of boomer in underground in this ML area.

Details are as below;

##### U/G development

Size of face : 3.0 m x 3.5 m (for mechanized development)

2.1M x 1.8 M (for manual operation)

Drilling pattern Wedge cut.& Parallel Cut Spacing : 0.60 M

Burden: 0.50 M Length of hole : 1.20 M

No.of holes : 16 Nos.

Explosives : 60 & 80%, 25mm dia.

##### U/G Mining in stope

Length of hole: 1.00 M

Spacing : 0.75 M

Burden : 0.60 M

Explosives : 60&80%, 25mm dia.

Powder factor : 7.98 ROM/KG.

### Magazine:

There is explosive magazine owned by MOIL Limited for Gumagon Mine of M/s MOIL Limited, which is located in 85.90 ha lease area. Services of this magazine will be used for this lease lead area for supply and storage of explosives. Details of magazine are as below; Gumgaon Magazine license No. is E/HQ/MH/22/483 (E-33527) Valid up to 31/03/2019.

The capacity of Magazine is:

|     |                    |   |             |
|-----|--------------------|---|-------------|
| (1) | Class 2            | : | 1000 Kgs.   |
| (2) | Class 6 Division 2 | : | 18750 Mtrs. |
| (3) | Class 6 Division 3 | : | 12500 Nos.  |

### Development:

- Stope blocks will be developed by driving Ore Drive and connecting it by X-cut of 3.0 x 3.5 m size.
- Winzes will be sunk from upper level to lower level.

### Ventilation System

After development of haulage road in -500'L in 6th Year at Ch.6600 vertical ventilation shaft of 3x3 m dia will be sunk from surface to 500'L and suitable ventilation fan will be installed after construction of evasee. Location has been marked on the surface plan.

The main intake is through vertical shaft and return is through Adit at 13'L.

One main fan of 75 HP is installed at 13'L for Ventilation of -100'L U/G and -200' L, - 300' L, and workings. The details of fan are as under:

Type : MAYA FAN

Capacity : 2820 Cu mtr/Min

Motor capacity : 75 H.P.

Water Gauge : 50 mm

Ventilation is checked periodically in the underground by using following instruments like

1. Anemometer
2. Hygrometer
3. Testo 410/2 (for air velocity with ambient temperature and humidity)
4. Multi gas detector( Gas Alert Microclip XT) Duct

pipes are also used in the underground to provide better ventilation to the workers at faces.

### **Proposed Travelling and Haulage Road**

#### **A. Traveling Roadways**

- Traveling roadways of dimension 3.6m (width) X 2.4 m (height) are always kept free from any obstacles.
- Every stope is connected with manway from lower level and a winze from upper level. Man ways are equipped with proper sized ladder, collars and kept always clean.
- Condition of roadways is regularly checked and sides and roof are properly dressed and kept free from loose. Height of the traveling ways is kept 1.8m.

#### **B. Ladder ways**

- Ladder ways of 1.5m dia in the stope are constructed with steel plate and maintained as per the provisions of MMR.
- collars have been provided at the interval of not more than 4.0 m and regular maintenance is being done.
- At the top of the ladder way the plates are raised to a height of one meter and top is covered and back is supported by a chock.

#### **C. Haulage Roadways**

- 30 lbs rails are being used for tramline, which are now being replaced by 50 lbs rails in phased manner.
- Standard iron sleepers are being used while laying the tramline. It is also seen that the concrete sleepers provide the encouraging result in loco transport.
- Rail joints are supported at both sides by providing the sleepers.
- Proper curves are being given wherever necessary, giving proper super elevation.

#### **D. Man Holes**

- Manholes have been provided in haulage road at stipulated interval.
- Sizes of the manholes are being kept as per statute. Manholes are periodically whitewashed and kept clean and free from any obstacles.

**(vi) Raw material required along with estimated quantity, likely source, marketing area of final product/s, Mode of transport of raw Material and Finished Products.**

The clean ore obtained after sorting of ROM will be stacked separately in stacking yard of lease of Gumgaon Mine which is 85.90 ha, from where the same material will be transported to customers by truck after measurement in weigh bridge commissioned at check point at Gumgaon Mine.

**(vii) Resource optimization/recycling and reuse envisaged in the project, if any, should be briefly outlined.**

Minerals are depleting asset once mined, they cannot be replenished like agriculture, vegetation thus a scientific approach will be taken up in exploitation of mineral with systematic method.

**(viii) Availability of water its source, energy/power requirement and source should be given.**

Ground water required through underground mining seepage (510 KLD) and bore well (210 KLD), total water requirement for various purpose will be 710 KLD.

| WATER REQUIREMENT |                                  |                 |
|-------------------|----------------------------------|-----------------|
| S.No              | Activities                       | Requirement KLD |
| 1                 | Domestic purpose                 | 200             |
| 2                 | Dust Suppression                 | 30              |
| 3                 | Green Belt Development           | 30              |
| 4                 | Agriculture                      | 400             |
| 5                 | Mining Operation (Sand, Stowing) | 50              |
| Total Requirement |                                  | 710             |

(Source: CGWA Application)

The power requirement will be met from M.S.E.B. 11 KV Line is provided up to village Khapa and near manganese deposit of Gumgaon area.

**(ix) Manpower Requirement**

Manpower:

|               |            |
|---------------|------------|
| U/G Workers   | 272        |
| Staff         | 164        |
| Officers      | 20         |
| <b>Total:</b> | <b>456</b> |

(x) **Quantity of wastes to be generated (liquid and solid) and scheme for their management/ disposal.**

There are 2 sources of waste generation in this ML area, development of drifts in rock and vertical deepening of ventilation shaft. The total quantum estimated in the 126.84 ha area is given below. There will not be any waste generation in first 5 years.

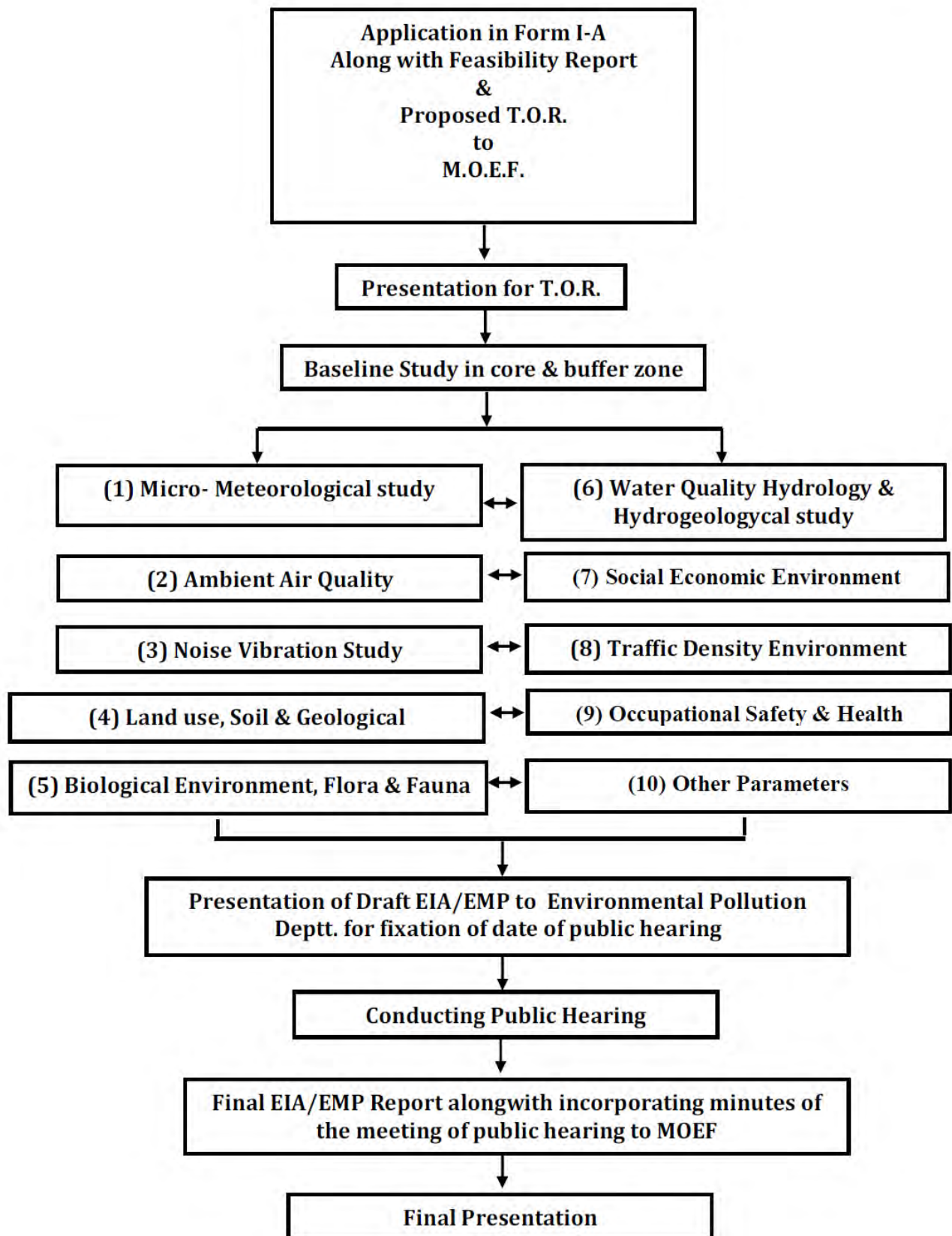
**Year wise total development in m<sup>3</sup> is appended in below table-**

| Year  | Footwall Haulage Road |                | X-cut |                | Total          |
|-------|-----------------------|----------------|-------|----------------|----------------|
|       | RMT                   | m <sup>3</sup> | RMT   | m <sup>3</sup> | m <sup>3</sup> |
| 1st   | 00                    | 00             | 00    | 00             | 00             |
| 2nd   | 00                    | 00             | 00    | 00             | 00             |
| 3rd   | 00                    | 00             | 00    | 00             | 00             |
| 4th   | 00                    | 00             | 00    | 00             | 00             |
| 5th   | 00                    | 00             | 00    | 00             | 00             |
| Total | 00                    | 00             | 00    | 00             | 00             |

generation of the waste during 1st 5 year of mining plan.

(xi) **Schematic representations of the feasibility drawing which give information of EIA purpose.**

As per the Environment Impact Assessment (EIA) notification dated 14th Sept. 2006, the proposed project falls under **Category 'A', Project Activity – 1(a)**. The EIA report is required to get environmental clearance for the project from the MoEF&CC the baseline studies will be undertaken as per Schematic diagram.



## 4.0 SITE ANALYSIS

### (i) Connectivity

| S.No. | Particulars       | Name                                                                                                                           | Distance & Directions |
|-------|-------------------|--------------------------------------------------------------------------------------------------------------------------------|-----------------------|
| 1.    | Nearest Village   | Gumgaon,                                                                                                                       | 0.31 Km. SE direction |
|       |                   | Khodegaon                                                                                                                      | 0.86 Km. NW direction |
|       |                   | Tegai                                                                                                                          | 0.3 Km S direction    |
| 2.    | Nearest town      | Khapa                                                                                                                          | 1 km. N direction     |
| 3.    | Nearest Highway   | NH-69                                                                                                                          | 5.20 km SW direction  |
|       |                   | SH-249                                                                                                                         | 0.70 km W direction   |
| 4.    | Nearest Railway   | Savner Railway Station                                                                                                         | 4.0 kms SW direction  |
| 5.    | Nearest airport   | Nagpur                                                                                                                         | 46.4 kms S direction  |
| 6.    | Nearest Forest    | ➤ Sitagondi– 6.6 Km in NE direction<br>➤ Khara Padri RF – 10.38 Km. in NW direction                                            |                       |
| 7.    | River/ Water body | ➤ Kanhar river – 0.78 Km. in E direction<br>➤ Kolar river – 3.1 Km. in SW direction<br>➤ Bhagimaheri – 12.9 Km in NE direction |                       |

### (ii) Land Form, Land use and Land ownership

Details of the lease 126.84 ha area for which the mining plan is submitted for approval is given below-

Land use pattern of M.L area –

| Total area in hect | Area under Revenue in hect | Area under private ownership in hect. |
|--------------------|----------------------------|---------------------------------------|
| 126.84             | 12.64                      | 114.20                                |

Details of the land:

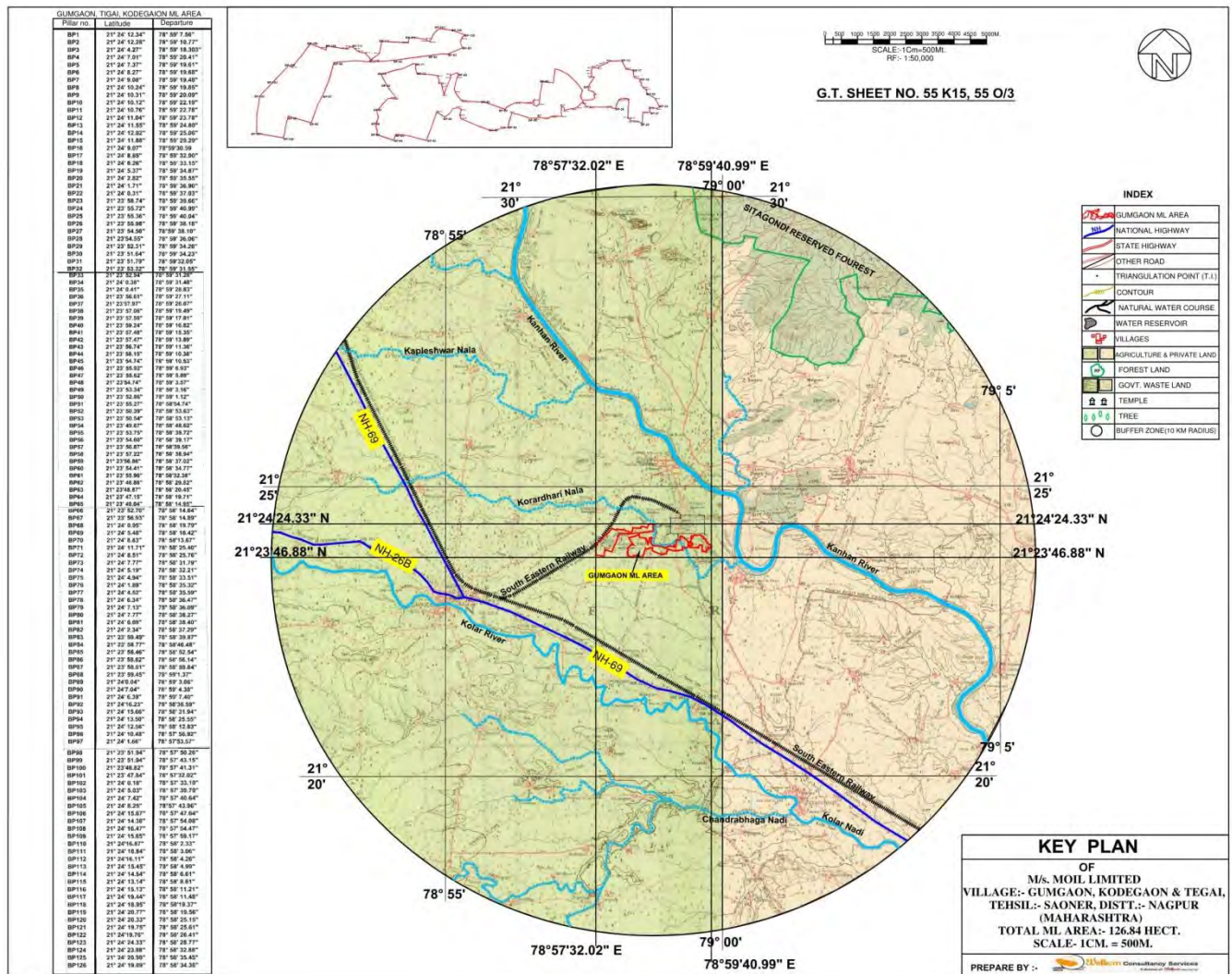
| Forest Land | Area (Ha) | Non Forest       | Area in ha |
|-------------|-----------|------------------|------------|
| NIL         | NIL       | Waste Land       | 15.90      |
|             |           | Grazing Land     | 21.19      |
|             |           | Agriculture land | 89.75      |
|             |           | Others           | -          |
|             |           | Residential Area | -          |
|             |           | Nalla            | -          |
|             |           | Total            | 126.84     |

**Total Lease area – 126.84 ha**

### (iii) Topography

The area is undulating with gently slope due East. The general MRL is 310 and the highest MRL recorded is 331 MRL to the west. The hillock is standing in E-W direction, on North part of the area, made up of quartzite of Sitasongi formation. A Korardhari Nala, which is a tributary of river Kanhan flows along the East limit of the area.

### Key Plan



### Drainage

The main water course of the area is Kanhan River which flows about 1.5 kms on the eastern side of the lease. The Korardhari nala flows through south of the lease hold area and join at 2.5 km down south-east to Kanhan River. Also there are number of non-perennial water courses and nalas originating from the above mentioned ridges from higher slopes of the hill to main drainage system of Kanhan River.

- (iv) Existing land use pattern (agriculture, non-agriculture, forest, water bodies (including area under CRZ), shortest distances from the periphery of the project to periphery of the forests, national area, wild life sanctuary, eco sensitive areas, water bodies (distance from the HFL of the river), CRZ. In case of notified industrial area, a copy of the Gazette notification should be industrial area, a copy of the Gazette notification should be given.

Existing land use pattern for 126.84 ha is given below:-

| Sr. No | Head                 | Area put on use at start of plan | Additional requireme nt during plan | Area considered as fully reclaimed & | Net area considered for calculation |
|--------|----------------------|----------------------------------|-------------------------------------|--------------------------------------|-------------------------------------|
| 1      | Area to be excavated | NIL                              | 1.00                                | Nil                                  | 1.00                                |
| 2.     | Storage for top soil | Nil                              | 0.25                                | Nil                                  | 0.25                                |
| 3.     | Overburden           | Nil                              | 0.25                                | Nil                                  | 0.25                                |
| 4.     | Mineral storage      | Nil                              | NIL                                 | Nil                                  | 0.00                                |
| 5.     | Infrastructure       | Nil                              | 0.25                                | Nil                                  | 0.25                                |
| 6      | Roads                | Nil                              | 0.25                                | Nil                                  | 0.25                                |
| 7      | Railways             | -                                | -                                   | -                                    |                                     |
| 8.     | Green Belt           | Nil                              | 5.00                                | 2.00                                 | 7.00                                |
| 9.     | Tailing pond         | -                                | -                                   | -                                    | -                                   |
| 10     | Effluent treatment   | -                                | -                                   | -                                    | -                                   |
| 11     | Mineral separation   | -                                | -                                   | -                                    | -                                   |
| 12     | Township area        | -                                | Nil                                 | Nil                                  | Nil                                 |
| 13     | Other to specify     | -                                | -                                   | -                                    | -                                   |
|        |                      |                                  |                                     | Total area                           | 9.00 Ha                             |

#### (v) Existing Infrastructure

##### A. Hospital:-

The mine has got a small hospital which is supervised by full time 2 medical officers and staff. It is also equipped with ambulance. Hospital services are available for 24 hrs. Details of the hospital and staff are given below;

Medical :

|    |                                     |    |
|----|-------------------------------------|----|
| 1. | Manager (M.S.)                      | 02 |
| 2. | Compounder                          | 02 |
| 3. | Nurses                              | 02 |
| 4. | No.of Beds available in female ward | 02 |
| 5. | No.of Beds available in male ward.  | 02 |

##### B. Canteen:

The canteen is situated near the Mine office of Gumgaon Mine. Canteen serves tea and snacks to the employees at subsidized rate.

### C. Drinking Water:

Drinking water is supplied from the overhead tank to the employees of the MOIL at Gumgaon Mine. There are other two dug-well near the residential colonies. 5 HP submersible pumps are installed at dug wells and in bore hole.

### D. Social Club:

This is provided for entertainment of employees of the mine. Social club has one colour TV set, periodicals and books. Various sports materials like Table tennis, Carrom boards are also provided.

### E. Housing :

Officers quarter available 30  
Staff quarter available 150  
Labour quarter available 200

### (vi) Soil

That Loose Sandy Soil is available here and there. The top soil will be stored separately in top soil dumps. This top soil will be used for plantation of new sapling on hill side and it will spread wherever sapling will be planted.

### (vii) Climatic Conditions

The prevailing climatic conditions were studied using following instruments.

|    |                   |                                 |
|----|-------------------|---------------------------------|
| 1. | Temperature       | Mercury thermometer             |
| 2. | Relative Humidity | Wet and dry bulb<br>Thermometer |
| 3. | Wind speed        | Anemometer                      |
| 4. | Wind direction    | Wind vane.                      |
| 5. | Cloudiness        | Visual inspections.             |

Results of the climate features observed are presented below;

## RESULT OF CLIMATIC OBSERVATIONS – 2015-16

| S. NO | Description                   | Summer | Monsoon | Winter |
|-------|-------------------------------|--------|---------|--------|
| 1.    | Predomination wind directions | SW     | SW      | NE     |
| 2.    | Predominant wind speed km/hr  | 11     | 10.0    | 11     |
| 3.    | Maximum temperature           | 44     | 35      | 25     |
| 4.    | Maximum temperature           | 26.5   | 24.5    | 12.5   |
| 5.    | Maximum R.H. %                | 77     | 92      | 58     |
| 6.    | Maximum R.H. %                | 38     | 68      | 31     |

### (viii) Social Infrastructure Available

#### a) Electricity Supply:

M.S.E.B. 11 KV Line is provided up to village Khapa and near manganese deposit of Gumgaon area.

#### b) Water Supply:

Drinking water is drawn from dug wells & tube wells & being supply to existing colony at Gumgaon Mine of MOIL.

#### c) Railway Siding:

The lease area has no railway siding. The Manganese from the lease area will be transported after measurement at Weigh Bridge at Gumgaon Mine of MOIL by trucks.

#### d) Roads:

The National Highway No. 69 is 5.20 Km SW & SH-249 – 0.70 km W direction from mine, which is connected by all weather roads.

#### e) Communication:

The communication is established by Bus Run by Maharashtra State Transport from Nagpur to Khapa and Ramtek. Telephone facility is being provided by Bharat Sanchar Nigam Limited.

#### f) Postal facilities:

Postal facilities is available at Khapa village which is 1 Km. away from the lease area. .

#### g) Schooling facilities:

Schooling facilities are available at khapa Where Primary/High schools are available.

#### h) Hospital:-

Company runs a Small Hospital at Gumgaon Mine which is 1 km from the lease area having two bed for male and two bed for female patients. Govt. Government hospital is available at Khapa and saoner.

## 5.0 PLANNING BRIEF

### (i) Planning Concept (type of industries, facilities, transportation etc.) Town and country Planning/Development authority Classification

This is a new proposed Manganese Mine. The project activities will come into operation stage after getting statutory clearances from concerned authorities. Further project activities will follow the applicable rules and regulation and provide the control measure and devices to achieve the standard norms. The mine will be operated by Underground method of mining. Maximum production of Manganese ore according to current approved Mine Plan will be 1,02,000 tonnes per annum (ROM). Mining is proposed in 126.84 Ha. mine lease area.

### (ii) Population Projection

The project will employ most of the workers from nearby areas. Local people from nearby villages will be give preference.

### (iii) Assessment of Infrastructure Demand (Physical & Social)

Demand of safety equipment, shelter and first aid room etc though many of facilities will soon established.

Demand for help in region of function education, social upliftment, Health etc.

### (iv) Amenities/Facilities

Communication facilities such as post office, telecommunications are available in near village. Nearby villages are well electrified. Medical facilities are available.

## 6.0 PROPOSED INFRASTRUCTURE

### (i) Industrial Area (Processing Area)

No processing unit is required

### (ii) Resident Area (Non Processing Area)

None

### (iii) Green Belt

The surrounding area is purely agricultural land with some waste land and grazing land round the area. It is proposed to put around 2000 sapling per year in hill side and where the low lying area, where the nallah is flowing culverts will be constructed.

Proposed afforestation in this area.

In this lease area the natural plant communities are not very well defined. MOIL will be undertaken afforestation work over 5.00 ha since 1st year by planting 2000

plant species per year. The details of proposed afforestation work since 1st year to 5th year is given below:

**Proposed area under plantation:**

| Year   | Area Of Plantation (Hectares) | No. Of Plants Planted |
|--------|-------------------------------|-----------------------|
| 1 year | 1.00                          | 2000                  |
| 2 year | 1.00                          | 2000                  |
| 3 year | 1.00                          | 2000                  |
| 4 year | 1.00                          | 2000                  |
| 5 year | 1.00                          | 2000                  |
| Total  | 5.00                          | 10000                 |

**PROPOSED AREA UNDER PLANTATION**

| Year             | Area (Ha) | Total Plantation |
|------------------|-----------|------------------|
| 1st to 5th Year  | 5.00      | 10000            |
| 6th to 10th Year | 2.00      | 4000             |
| Total            | 7.00      | 14000            |

2000 sapling will be done per year and achieve target of 33%.

**(iv) Corporate Social Responsibility (CSR)**

Corporate Social Responsibility (CSR) activities are developed as per the CSR policy of the MOIL Ltd.

**(v) Connectivity (Traffic and transportation Road/Rail/Metro/Water ways etc.)**

The project site is located about 0.7 km towards SE of village Khapa. Nearest railway station is Savner Railway Station (5.0 kms SW) from the project site and NH-69 is 5.20 km SW direction & SH-249 is 0.70 km W direction from mine site.

**(vi) Drinking Water Management (Source & supply of water)**

Drinking water is supplied from the overhead tank to the employees of the MOIL at Gumgaon Mine. There are other two dug-well near the residential colonies. 5 HP submersible pumps are installed at dug wells and in bore hole.

**(vii) Industrial Waste Management**

Not required

**(viii) Solid Waste Management**

If any waste generates, it will be secured over the area marked on plan. It is proposed to store the quantity with single layer of 5 mts height, 100 mts length &

20mts width with 30° slope. On maturity of dump capacity, it will be stabilized by plantation. These dumps will be stacked/located in main lease of 126.84 ha.

## 7.0 REHABILITATION & RESETTLEMENT (R&R PLAN)

**(Policy to adopted (Central/State) in respect of the project affected person including home oustees, land oustee and landless labour (A brief outline to be given)).**

There is no hutment in the lease area. No human being displaced from the area so no person will be affected contrary local people will get job opportunity.

## 8.0 PROJECT SCHEDULE AND COST ESTIMATES

i. **(Likely date of start of construction and likely date of completion (time schedule for the project to be given))**

Project will commence within one month after getting environmental clearance from MoEF, Delhi.

ii. **Estimated project cost along with analysis in terms of economic viability of the project**

Total Project Cost : Rs. 100 Crores

## 9.0 ANALYSIS OF PROPOSAL (FINAL RECOMMENDATIONS)

**Financial and social benefits with special emphasis on the benefit to the local people including tribal population, if any, in the area.**

There is no industry in the area. The mining in this area will provide employment to nearby villagers. This employment opportunity will help in raising the standard of living of the people in the area. Socio-economic condition of area generally improve as mining activities create additional employment opportunity for local habitant, social – economic status of local populace also improves as a result of social welfare activities under taken by mine owners. Significant contribution will be made towards education medical facilities and cultural aspects. Drinking water facilities will be provided to villagers, plantation will improve environmental condition of the area.

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